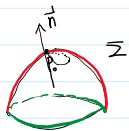


作业: P81. 1, 2, 3, 4, 5, 6, 8, 9, 10.

1. 曲面定向.

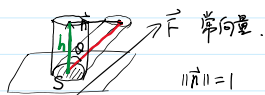


双侧曲面.

曲面正侧: 双侧曲面正法向量指向一侧. $\Sigma^+ \text{ —— } \Sigma \text{ 正侧}$ $\Sigma^- \text{ —— } \Sigma \text{ 负侧}$

有向曲面:

2. 物理来源.

设 $\Omega \subset \mathbb{R}^3$ 内有某种流体稳定流动. (与时间无关, 质量均匀分布).在 $\forall M(x, y, z) \in \Omega$ 流速 $\vec{F}(M) = (P(M), Q(M), R(M))$. $P, Q, R \in C(\Omega)$.设在 Ω 内有一片光滑双侧曲面 Σ . 在单位时间, 流体流过 Σ 体积, 即为流体流过 Σ 流量.计算: 在单位时间内, 流体流过 Σ 流量.

$$\vec{F} \cdot \vec{n} = \|\vec{F}\| \cos \theta = h$$

$$Q = Sh = \vec{F} \cdot \vec{n} S$$

$$\forall T = \{\Sigma_1, \Sigma_2, \dots, \Sigma_n\}.$$

$$\|T\| = \max \{ \text{diam } \Sigma_i \mid i=1, 2, \dots, n \}.$$

$$\forall M_i \in \Sigma_i.$$

$$\Sigma_i \text{ 在 } M_i \text{ 法向量 } \vec{n}_i, \|\vec{n}_i\|=1.$$

$$Q = \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n \vec{F}(M_i) \cdot \vec{n}_i \Delta S_i$$

$$\Sigma \text{ 面积 } \Delta S_i.$$

$$\vec{n}_i = (\cos \alpha_i, \cos \beta_i, \cos \gamma_i)$$

$$\vec{n}_i \Delta S_i = (\cos \alpha_i \Delta S_i, \cos \beta_i \Delta S_i, \cos \gamma_i \Delta S_i)$$



$$\Delta S_i \cos \gamma_i = \pm \Delta x_i \Delta y_i = \Delta x_i \wedge \Delta y_i$$

 Σ_i 投影在 Oxy 平面上有向面积.

$$\Delta S_i \cos \alpha_i = \Delta y_i \wedge \Delta z_i$$

$$\Delta S_i \cos \beta_i = \Delta z_i \wedge \Delta x_i$$

$$\lim_{\|T\| \rightarrow 0} \sum_{i=1}^n P(M_i) \Delta y_i \wedge \Delta z_i + Q(M_i) \Delta z_i \wedge \Delta x_i + R(M_i) \Delta x_i \wedge \Delta y_i.$$

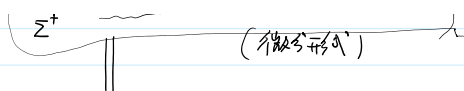
定义1. 设 $\vec{F}(M) = (P(M), Q(M), R(M))$ 定义在光滑双侧曲面 Σ 上.在 Σ 指定一侧分割. $T = \{\Sigma_1, \Sigma_2, \dots, \Sigma_n\}$. $\|T\| = \max \{ \text{diam } \Sigma_i \mid i=1, 2, \dots, n \}$. $\Delta y_i \wedge \Delta z_i, \Delta z_i \wedge \Delta x_i, \Delta x_i \wedge \Delta y_i$ 分别表示 Σ_i 在 Oyz, Oyz, Oxy 平面上投影有向面积.

$$\forall M_i \in \Sigma_i$$

若 $\lim_{\|T\| \rightarrow 0} \sum_{i=1}^n P(M_i) \Delta y_i \wedge \Delta z_i + Q(M_i) \Delta z_i \wedge \Delta x_i + R(M_i) \Delta x_i \wedge \Delta y_i$ 存在且划分方式 T 无关. 与点取法. 则称此极限为 $\vec{F}$ 沿 Σ^+ 第II型曲面积分.

$$\iint_{\Sigma^+} P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy.$$

(微分形式)



$$\iint_{\Sigma^+} (P, Q, R) \cdot \underbrace{(dy \wedge dz, dz \wedge dx, dx \wedge dy)}_{\substack{\parallel \\ \vec{n} dS \text{ — 有向面积, 微元}}} = \iint_{\Sigma^+} \vec{F} \cdot \underbrace{d\vec{S}}_{\text{有向面积'}}$$

$$\vec{n} = (\cos \alpha, \cos \beta, \cos \gamma)$$

$$\cos \alpha dS = dy \wedge dz$$

$$\cos \beta dS = dz \wedge dx$$

$$\cos \gamma dS = dx \wedge dy$$

$$\begin{aligned} \iint_{\Sigma^+} \vec{F} \cdot \vec{n} dS &= - \iint_{\Sigma^-} \vec{F} \cdot (-\vec{n}) dS \\ &= \iint_{\Sigma^+} (P, Q, R) \cdot \vec{n} dS \end{aligned} \quad \left\{ \begin{aligned} \vec{n} &= \frac{(-f'_x, -f'_y, 1)}{\sqrt{1+f_x'^2+f_y'^2}} \\ dS &= \sqrt{1+f_x'^2+f_y'^2} dx dy \end{aligned} \right.$$

$$\iint_{\Sigma} (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS \quad \text{—— 第 I 型}$$

$$\iint_{\Sigma} P \cos \alpha dS + \iint_{\Sigma} Q \cos \beta dS + \iint_{\Sigma} R \cos \gamma dS$$

$$\iint_{\Sigma^+} P dy \wedge dz + \iint_{\Sigma^+} Q dz \wedge dx + \iint_{\Sigma^+} R dx \wedge dy$$

$$\Sigma \text{ 定向} \quad \iint_{\Sigma^+} \vec{F} \cdot d\vec{S}$$

定理 2. 设 $\Sigma: z = f(x, y) \in C^1(D)$. $D \subset \mathbb{R}^2$ 有界闭区域.

$$\forall M(x, y, z) \in \Sigma, \quad \vec{n} = \frac{(-f'_x, -f'_y, 1)}{\sqrt{1+f_x'^2+f_y'^2}}, \quad \text{设 } P(x, y, z), Q(x, y, z), R(x, y, z) \in C(\Sigma).$$

$$\begin{aligned} \text{则 } & \iint_{\Sigma^+} P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy \\ &= \pm \iint_D [P(x, y, f(x, y)) \cdot (-f'_x) + Q(x, y, f(x, y)) \cdot (-f'_y) + R(x, y, f(x, y))] dx dy \end{aligned}$$

其中, $\pm$ 重积分前正, 负取决于法向量与 z 轴正向所成角是锐角、钝角

$$\text{定理 3. 设光滑 } \Sigma: \begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} \quad (u, v) \in D.$$

$$\forall (u, v) \in D, \quad \vec{n} = \frac{\pm(A, B, C)}{\sqrt{A^2+B^2+C^2}}, \quad A = \frac{D(y, z)}{D(u, v)}, \quad B = \frac{D(z, x)}{D(u, v)}, \quad C = \frac{D(x, y)}{D(u, v)}.$$

$$\text{设 } P(x, y, z), Q(x, y, z), R(x, y, z) \in C(\Sigma).$$

$$\begin{aligned} \text{则 } & \iint_{\Sigma^+} P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy \\ &= \pm \iint_D [P(u, v)A + Q(u, v)B + R(u, v)C] du dv = \pm \iint_D \begin{vmatrix} P & Q & R \\ x'_u & y'_u & z'_u \end{vmatrix} du dv \end{aligned}$$

Σ^+

$$= \pm \iint_D [P(u,v)A + Q(u,v)B + R(u,v)C] du dv = \pm \iint_D \begin{vmatrix} P & Q & R \\ x'_u & y'_u & z'_u \\ x'_v & y'_v & z'_v \end{vmatrix} du dv$$

其中, $\pm$ 取决于 $\vec{n}$ 符号.

例1. 计算 $I = \iint_{\Sigma^+} \frac{1}{x} dy dz + \frac{1}{y} dz dx + \frac{1}{z} dx dy$

Σ^+ : $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. 外侧为正. $\iint_{\Sigma^+} \frac{1}{x} dy dz + \frac{1}{y} dz dx + \frac{1}{z} dx dy$

解: $\vec{n} = \frac{(A, B, C)}{\sqrt{A^2+B^2+C^2}}$ Σ^+ : $\begin{cases} x = a \sin \varphi \cos \theta \\ y = b \sin \varphi \sin \theta \\ z = c \cos \varphi \end{cases} \quad \varphi \in [0, \pi], \theta \in [0, 2\pi]$

$$A = \frac{D(y, z)}{D(\varphi, \theta)} = \begin{vmatrix} b \cos \varphi \sin \theta & b \sin \varphi \cos \theta \\ -c \sin \varphi & 0 \end{vmatrix} = bc \sin^2 \varphi \cos \theta$$

$$B = \frac{D(z, x)}{D(\varphi, \theta)} = ac \sin^2 \varphi \sin \theta$$

$$C = \frac{D(x, y)}{D(\varphi, \theta)} = ab \sin \varphi \cos \varphi$$

$$\begin{aligned} \iint_{\Sigma^+} \frac{1}{z} dx dy &= \iint_{\substack{\theta \in [0, 2\pi] \\ \varphi \in [0, \pi]}} \frac{1}{c \cos \varphi} \cdot ab \sin \varphi \cos \varphi d\varphi d\theta \\ &= \int_0^{2\pi} d\theta \int_0^\pi \frac{ab}{c} \sin \varphi d\varphi = \frac{4\pi ab}{c} \end{aligned}$$

$$\iint_{\Sigma^+} \frac{1}{x} dy dz = \frac{4\pi bc}{a}$$

$$\iint_{\Sigma^+} \frac{1}{y} dz dx = \frac{4\pi ac}{b}$$

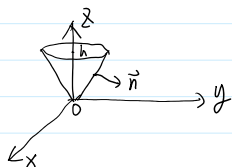
$$I = \frac{4\pi}{abc} (a^2b^2 + a^2c^2 + b^2c^2)$$

例2. 计算 $I = \iint_{\Sigma^+} (x^2 \cos \alpha + y^2 \cos \beta + z^2 \cos \gamma) dS$

其中 Σ^+ : $x^2 + y^2 = z^2$ ($0 \leq z \leq h$) 为外侧

$\vec{n} = (\cos \alpha, \cos \beta, \cos \gamma)$ 为 Σ^+ 单位法向量.

解:



$$\vec{n} = \frac{(z'_x, z'_y, -1)}{\sqrt{1+z'^2_x+z'^2_y}}$$

$$\Sigma^+: z = \sqrt{x^2 + y^2}, \quad x^2 + y^2 \leq h^2$$

$$I = \iint_{x^2+y^2 \leq h^2} (x^2, y^2, z^2) \cdot (z'_x, z'_y, -1) dx dy$$

$$z'_x = \frac{x}{\sqrt{x^2+y^2}}$$

$$z'_y = \frac{y}{\sqrt{x^2+y^2}}$$

$$= \iint_{x^2+y^2 \leq h^2} \left[\frac{x^3+y^3}{\sqrt{x^2+y^2}} - (x^2+y^2) \right] dx dy$$

$$= - \iint_{x^2+y^2 \leq h^2} (x^2+y^2) dx dy$$

$$x^2+y^2 \leq h^2 \quad x+y$$

$$= - \iint_{x^2+y^2 \leq h^2} (x^2+y^2) dx dy$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$= - \int_0^{2\pi} d\theta \int_0^h r^2 \cdot r dr = - \frac{\pi h^4}{2}$$

例13. 计算 $I = \iint_{\Sigma^+} (y-z) dy \wedge dz + (z-x) dz \wedge dx + (x-y) dx \wedge dy$

Σ^+ : $z = \sqrt{4x-x^2-y^2}$ 含在柱面 $x^2+y^2=2x$ 内部部分. 方向向下.

解: Σ^+ : $z = \sqrt{4x-x^2-y^2}$. $x^2+y^2 \leq 2x$. 方向向下.

$$\left((x-2)^2 + y^2 + z^2 = 4 \quad \vec{n} = -\frac{(x-2, y, z)}{2} \right)$$

$$I = - \iint_{\Sigma^+} (y-z, z-x, x-y) \cdot \frac{1}{2} (x-2, y, z) dS$$

$$= -\frac{1}{2} \iint_{\Sigma} [(y-z)(x-2) + (z-x)y + (x-y)z] dS$$

$$yx - zy - zx + 2z$$

$$+ zy - yx + xz - yz$$

$$= \iint_{\Sigma} (y-z) dS = \iint_{\Sigma} y dS - \iint_{\Sigma} z dS$$

$$= - \iint_{\Sigma} z dS = - \iint_{x^2+y^2 \leq 2x} \sqrt{4x-x^2-y^2} \cdot \sqrt{1+z_x'^2+z_y'^2} dx dy$$

$$z_x' = \frac{-(2-x)}{\sqrt{4x-x^2-y^2}}$$

$$z_y' = \frac{-y}{\sqrt{4x-x^2-y^2}}$$

$$= -2 \iint_{x^2+y^2 \leq 2x} dx dy = -2\pi$$

二. 高斯公式.

定理4. 设 $\Omega \subset \mathbb{R}^3$ 有界闭域. $\partial\Omega = \Sigma$ 是光滑或分片光滑封闭曲面.

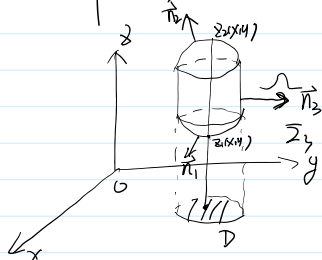
若 $P(x, y, z), Q(x, y, z), R(x, y, z) \in C^1(\Omega, \mathbb{R})$.

$$\oiint_{\Sigma^+} P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz$$

$$\oiint_{\Sigma^+} P dy \wedge dz + \oiint_{\Sigma^+} Q dz \wedge dx + \oiint_{\Sigma^+} R dx \wedge dy$$

$$\oiint_{\Sigma^+} R dx \wedge dy = \iiint_{\Omega} \frac{\partial R}{\partial z} dx dy dz = \iint_D dx dy \int_{z_1(x,y)}^{z_2(x,y)} \frac{\partial R}{\partial z} dz = \iint_D [R(x, y, z_2(x, y)) - R(x, y, z_1(x, y))] dx dy$$

$$I. \Omega = \{(x, y, z) \mid (x, y) \in D, z_1(x, y) \leq z \leq z_2(x, y)\}$$



$$z_1(x, y), z_2(x, y) \in C^1(D)$$

$$\vec{n}_1 = \frac{(z_1', z_1'', -1)}{\sqrt{1+z_1'^2+z_1''^2}}, \quad \vec{n}_2 = \frac{(-z_2', -z_2'', 1)}{\sqrt{1+z_2'^2+z_2''^2}}$$

↙_x

$\frac{1}{D}$

$\sqrt{1 + z_{1x}^2 + z_{1y}^2}$

$\sqrt{1 + z_{2x}^2 + z_{2y}^2}$

$$\begin{aligned} \iint_{\Sigma^+} R dx \wedge dy &= \iint_{\Sigma_1^+} R dx \wedge dy + \iint_{\Sigma_2^+} R dx \wedge dy + \iint_{\Sigma_3^+} R dx \wedge dy \\ &= - \iint_D R(x, y, z_1(x, y)) dx dy + \iint_D R(x, y, z_2(x, y)) dx dy \end{aligned}$$

